

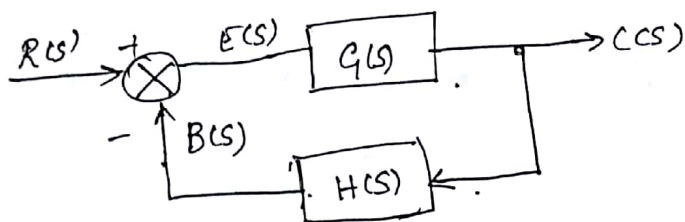
Signal Flow Graphs (SFG)

①

Block diagrams are very useful for representing control systems, but for complicated systems, the block diagram reduction process is tedious and time consuming. An alternate approach is that of SFG, developed by S.T. Mason, which does not need any reduction process because of the availability of a flow graph gain formula, which relates the i/p & o/p system variables.

A SFG is a graphical representation of the relationships b/w the variables of a set of linear algebraic equations. It consists of a network in which nodes representing each of the system variables are connected by directed branch.

consider a closed loop system.



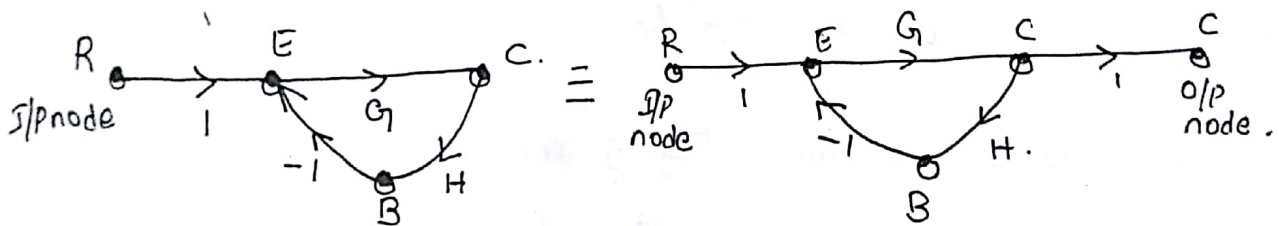
$R(s)$ = Reference Input.

$C(s)$ = o/p signal.

$B(s)$ = feedback signal.

$E(s)$ = Actuating signal
 $= R(s) - B(s)$

The above B.D can be represented in terms of SFG as.



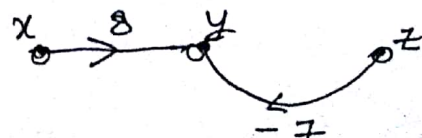
A. SFG is used to represent.

- (a) Set of linear algebraic equations.
- (b) A block diagram.
- (c) A network.

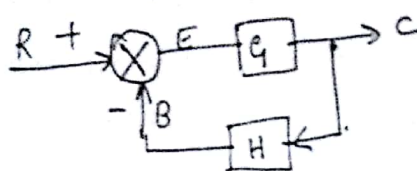
(a) Algebraic Equations

$$Y = 8X - 7Z$$

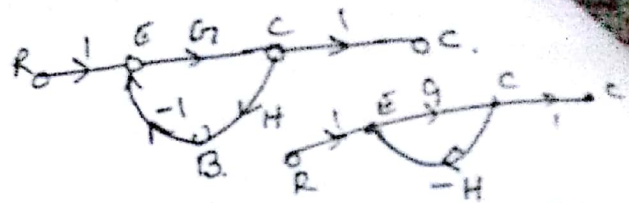
SFG Representation.



⑥ Block diagram.



SFG.

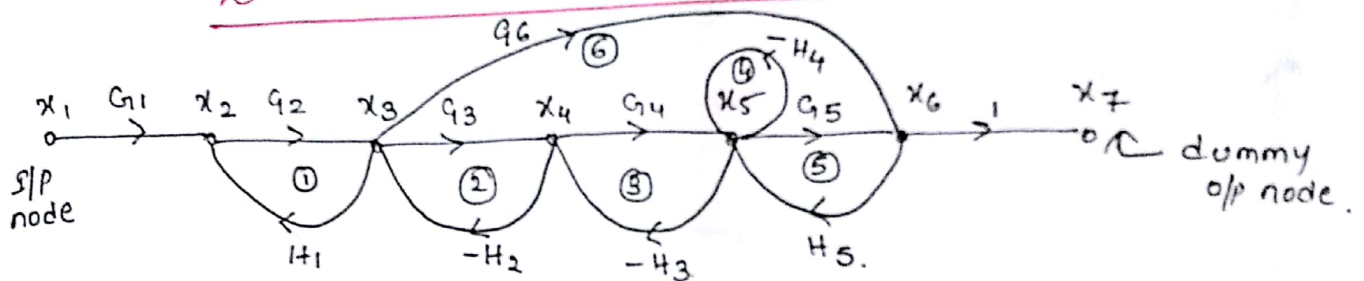


Note: If a summing pt & take off pt are near each other ~~then~~ in a B.D., they are represented by separate rules in the ^{corresponding} SFG.

⑦ Network:

If a network is given, then equations are written & then SFG is drawn as shown in Eqn (a)

Definitions Associated with SFG



① Input node: It is a node which has outgoing branches only (x_1)

② Output node: It is a node which has incoming branches only (x_7).

In case of an output node having an outgoing branch, then it is possible to extend the SFG through a branch of unity gain to another node called dummy node (x_7). This process does not alter T.F.

③ Forward Path: It is the path traced from i/p to the o/p in the direction of the signal, without passing through any node more than once.

Eg: In the above fig, the forward paths are.

(M₁) ④ $x_1 x_2 x_3 x_4 x_5 x_6 x_7$ (M₂) ⑤ $x_1 x_2 x_3 x_6 x_7$

④ Forward path gain (M): It is the product of all the gains or transmittance along the forward path.

eg: There are 2 forward paths & hence the forward path gains.

① $M_1 = G_1 G_2 G_3 G_4 G_5$ ② $M_2 = G_1 G_2 G_6$

⑤ closed loop: It is the path traced by the signal from node and back to it.

It is a path which originates and terminates at the same node, without tracing any node more than once.

⑥ loop gain: It is the product of all the gains along the closed loop.

sl no.	Identity for closed loop	loop gain
1	$X_2 X_3 X_2$	$G_2 H_1$
2	$X_3 X_4 X_3$	$-G_3 H_2$
3	$X_4 X_5 X_4$	$-G_4 H_3$
4	$X_5 X_6 X_5$	$G_5 H_5$
5	$X_5 X_5$	$-H_4$
⑥	$X_3 X_4 X_5 X_3$	
6.	$X_3 X_6 X_5 X_4 X_3$	$G_6 H_5 H_3 H_2$

⑦ Self or delay loop: It is a closed loop involving only one node.

eg: loop. $X_5 X_5$ is a self loop with loop gain $-H_4$.

⑧ Non-touching loops: Loops are said to be non-touching if they don't possess any common node.

Sl. no.	Combination of two non touching loops	Gain product
1	1 & 3	$(G_2 H_1) (-G_4 H_3)$
2	1 & 4	$(G_2 H_1) (-H_4)$
3	1 & 5	$(G_2 H_1) (G_5 H_5)$
4	2 & 4	$(-G_3 H_2) (-H_4)$
5	2 & 5	$(-G_3 H_2) (G_5 H_5)$

note:

There are no combinations of 3 non-touching loops.

Mason's Gain Formula: This formula is useful in determining the overall P.F of a system from the SFG. It is given by the relation

$$\frac{C(s)}{R(s)} = T = \frac{\sum_{k=1}^n M_k \Delta_k}{\Delta} = \sum_k \frac{P_k \Delta_k}{\Delta}$$

where,

$\Delta_k = 1 - (\text{sum of the loop gains of individual loops not touching the } k^{\text{th}} \text{ forward path}) + (\text{sum of the gain products of all possible combinations of two non touching loops, which are not touching the } k^{\text{th}} \text{ forward path}) - \dots$

$k = \text{no. of forward paths} = n.$

$M_k = \text{Gain of } k^{\text{th}} \text{ forward path.}$

$\Delta_k = \text{The value of } \Delta \text{ for that part of graph not touching the } k^{\text{th}} \text{ forward path. or the value of } \Delta \text{ by eliminating all the loop gain \& \text{ associative products which are touching the } k^{\text{th}} \text{ forward path.}$

Δ = determinant of the graph.

(3)

= $1 - (\text{sum of loop gains of all individual loops})$

+ $(\text{sum of gain products of all possible combinations of two non touching loops})$

- $(\text{sum of gain products of all possible combinations of three non touching loops})$

+ $\dots$

ie. $\Delta = 1 - \sum_m P_{m1} + \sum_m P_{m2} - \sum_m P_{m3} + \dots$

P_{mr} = Gain product of m^{th} possible combination of r non touching loops.

For the above example the gain formula is.

$$T = \frac{X_7}{X_1} = \frac{X_6}{X_1} = \frac{M_1 \Delta_1 + M_2 \Delta_2}{1 - P_{M1} + P_{M2} - P_{M3} + \dots}$$

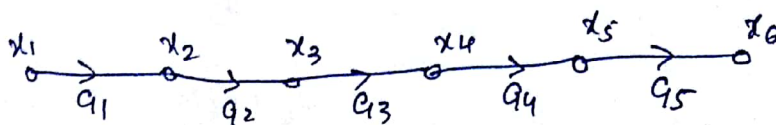
Already M_1, M_2, P_{M1} & P_{M2} are calculated.

To calculate Δ_1

* Select forward path $\rightarrow$ ①

* Erase all closed loops linked with forward path $\rightarrow$ ①.

* Also erase all closed loops of other forward paths, linked with Forward path $\rightarrow$ ①.



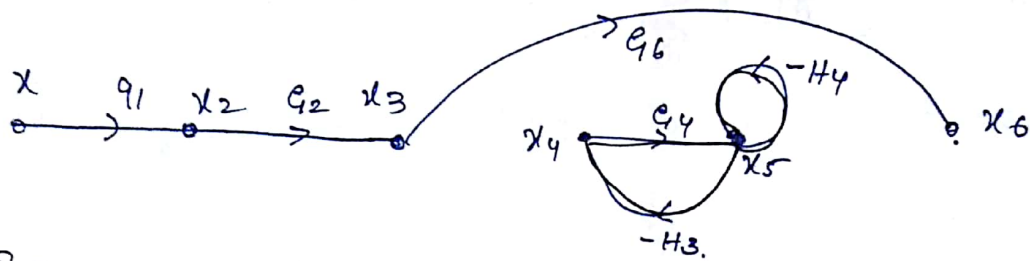
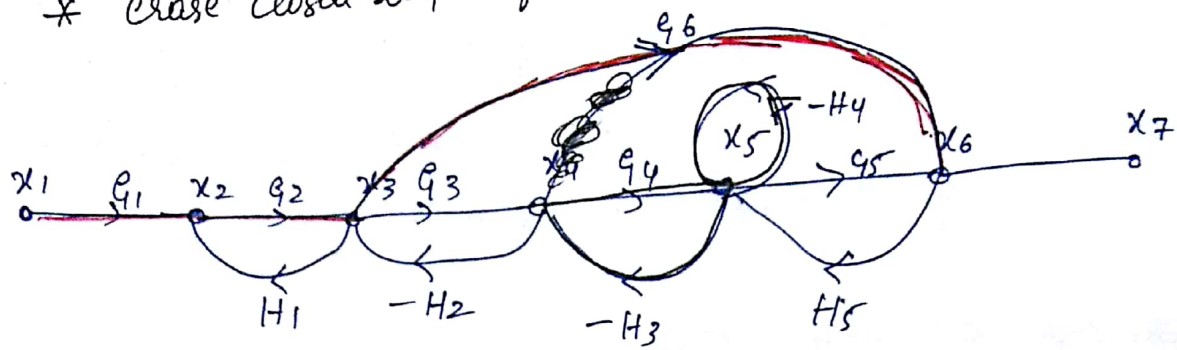
$$\Delta_1 = 1 - P_{M1} + P_{M2} = 1 - 0 + 0 = 1$$

$\therefore$ Forward path touches all the loops.

To calculate Δ_2 .

* Erase closed loops related to FP-②.

* Erase closed loops of other FP linked with FP-②.



There are two loops which are not touching FP-②

$$\therefore \Delta_2 = 1 - (-G_4 H_3 - H_4) + 0$$

$$= 1 - \text{sum of loop gains} + \text{sum of gains of two non touching loops} - \dots$$

Note: There are no two non touching loops which are not touching the 2nd forward path.

$$\Delta_2 = 1 + G_4 H_3 + H_4$$

Substituting these values in gain formula.

$$\frac{C(s)}{A(s)} = T = \frac{G_1 G_2 G_3 G_4 G_5 (1) + G_1 G_2 G_6 (1 + G_4 H_3 + H_4)}{1 - (G_2 H_1 - G_3 H_2 - G_4 H_3 + G_5 H_5 - H_4 + G_6 H_5 H_2 H_3) + (-G_2 G_4 H_1 H_3 - G_2 H_1 H_4 + G_2 H_5 H_1 G_5 + G_3 H_2 H_4 - G_3 H_3 G_5 H_5)}$$

4) construction of SFQ from algebraic equations

Consider a system described by the following set of equations

$$x_2 = a_{12} x_1 + a_{32} x_3 + a_{42} x_4 + a_{52} x_5 \rightarrow (1)$$

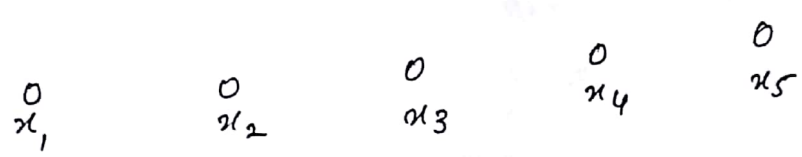
$$x_3 = a_{23} x_2 \rightarrow (2)$$

$$x_4 = a_{34} x_3 + a_{44} x_4 \rightarrow (3)$$

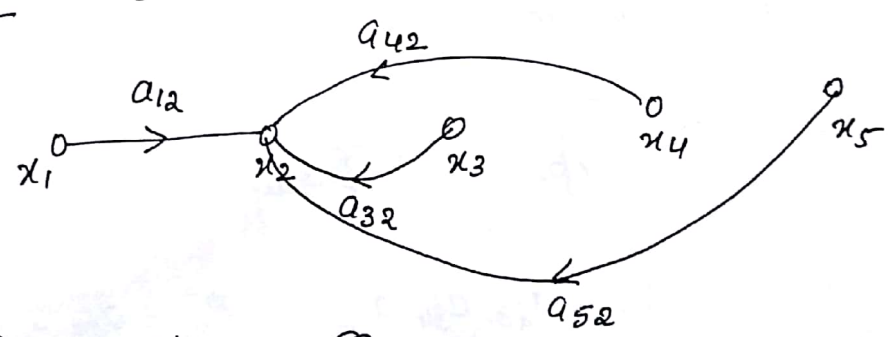
$$x_5 = a_{35} x_3 + a_{45} x_4 \rightarrow (4)$$

where x_1 is the i/p variable & x_5 is the o/p variable.

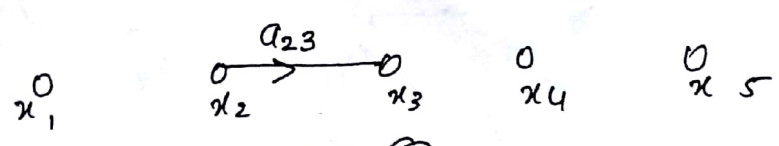
Step 1 locate the nodes as shown.



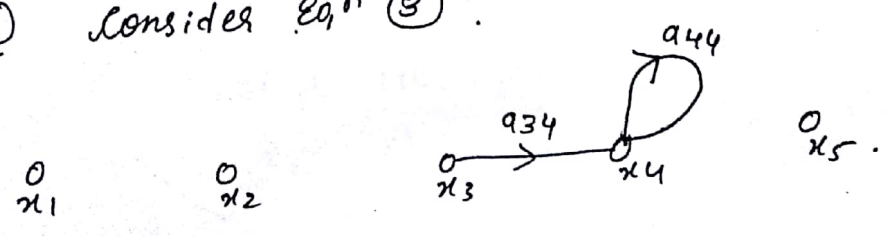
Step 2 consider the eqn (1).



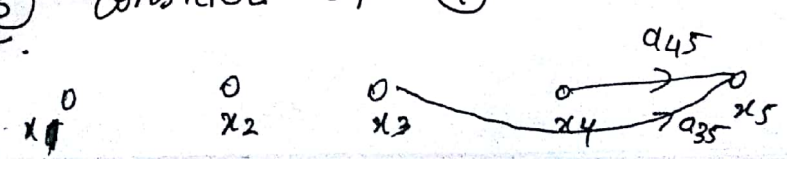
Step 3 consider eqn (2).



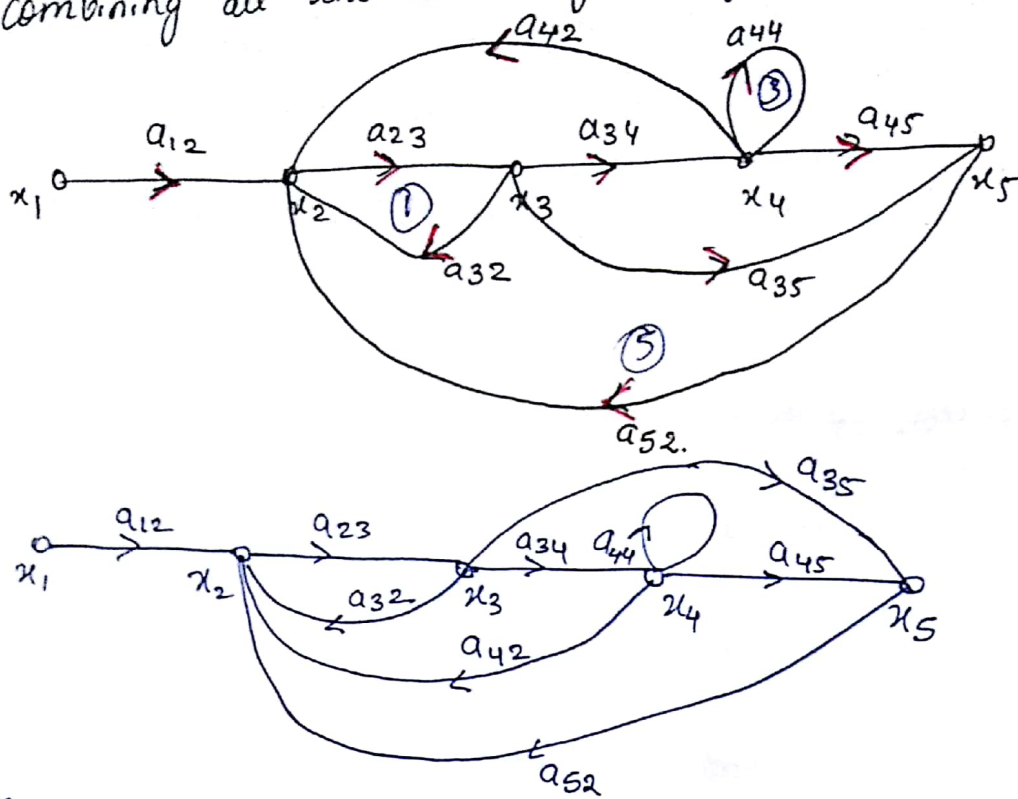
Step 4 consider eqn (3).



Step 5 consider eqn (4).



combining all the above figs we get overall SF9.



According to Mason's gain formula.

$$T = \frac{1}{\Delta} \sum M_k \Delta_k$$

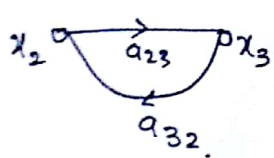
① There are 2 forward paths with path gains

$$M_1 = a_{12} a_{23} a_{34} a_{45}$$

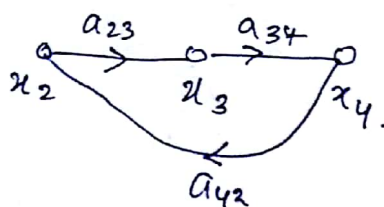
$$M_2 = a_{12} a_{23} a_{35}$$

② There are 5 individual loops with loop gain

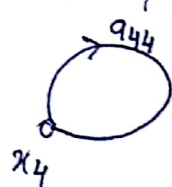
$$P_{11} = a_{23} a_{32}$$



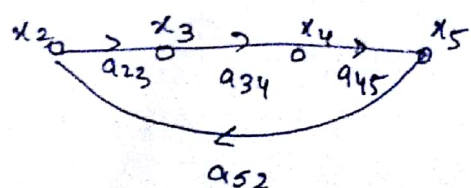
$$P_{21} = a_{23} a_{34} a_{42}$$



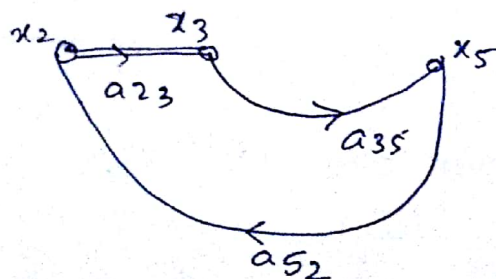
$$P_{31} = a_{44}$$



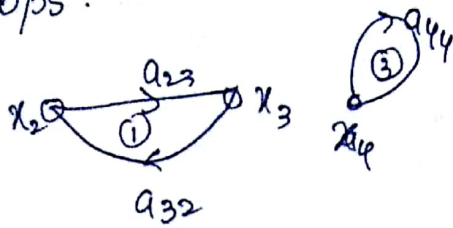
$$P_{41} = a_{23} a_{34} a_{45} a_{52}$$



$$P_{51} = a_{23} a_{35} a_{52}$$

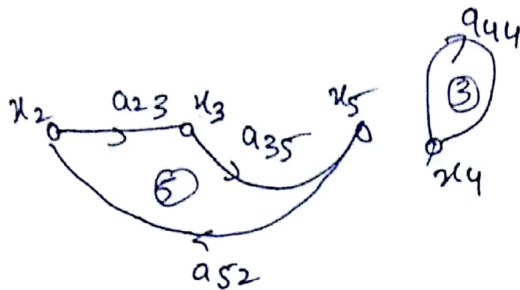


There are 2 possible combinations of two-non touching loops. (5)



1. 3 loops are not touching.

$$a_{23} a_{32} a_{44}$$



2. 3 loops are not touching.

$$a_{23} a_{35} a_{52} a_{44}$$

There are no possible combinations of three non touching, four non touching loops etc $\therefore P_{M3} = P_{M4} = \dots = 0$

$$\Delta = 1 - P_{M1} + P_{M2}$$

$$= 1 - (a_{23} a_{32} + a_{23} a_{34} a_{42} + a_{44} + a_{23} a_{34} a_{45} a_{52} + a_{23} a_{35} a_{52}) + (a_{23} a_{32} a_{44} + a_{23} a_{35} a_{52} a_{44})$$

$$T = \frac{M_1 \Delta_1 + M_2 \Delta_2}{\Delta}$$

$\Delta_1 = 1$, $\therefore$ first FP is in touch with all the loops.

$\Delta_2 = 1 - a_{44}$, $\therefore$ 2nd FP is in touch with all the loops except a_{44} loop.

③ Draw the SFG for the system eqn given & obtain the overall TF.

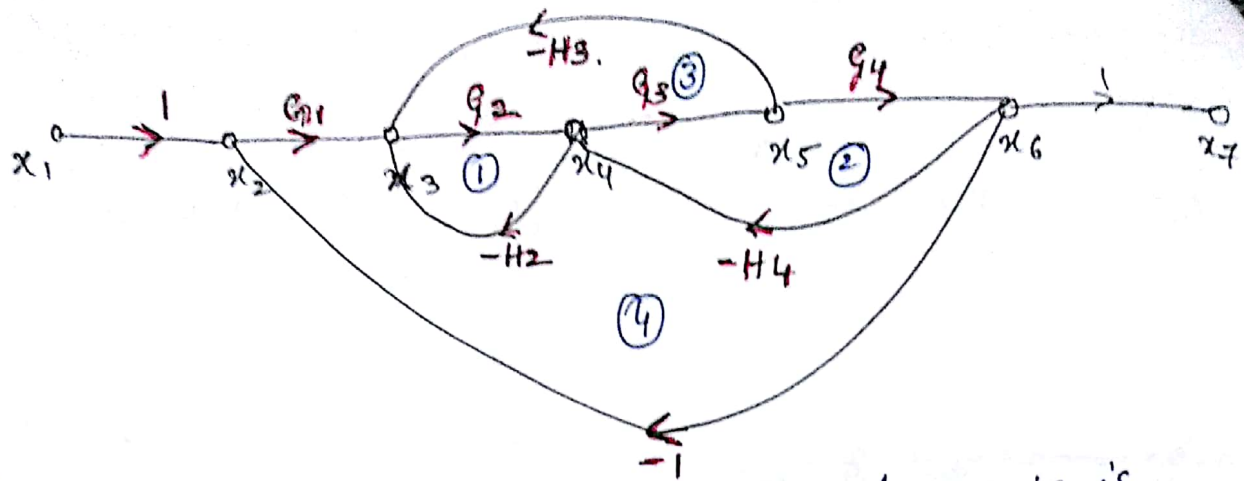
$$X_2 = X_1 - X_6$$

$$X_3 = G_1 X_2 - H_2 X_4 - H_3 X_5$$

$$X_4 = G_2 X_3 - X_6 H_4$$

$$X_5 = G_3 X_4$$

$$X_6 = G_4 X_5$$



① There is only one forward path whose gain is
 $M_1 = G_1 G_2 G_3 G_4$.

② There are 4-loops with loop gains.

$$P_{11} = -G_2 H_2 \quad P_{21} = -G_3 G_4 H_4 \quad P_{31} = -G_2 G_3 H_3 ;$$

$$P_{41} = -G_1 G_2 G_3 G_4$$

③ There are no possible combination of 2 non touching loops.
 $\therefore P_{M2} = 0$

④ $\Delta_1 = 1$, $\therefore$ the 1st FP is in touch with all the loops.

$$\therefore TF = \frac{M_1 \Delta_1}{\Delta} = \frac{G_1 G_2 G_3 G_4 (1)}{1 + (G_2 H_2 + G_3 G_4 H_4 + G_2 G_3 H_3 + G_1 G_2 G_3 G_4) + 0}$$

④ $x_1 = R - x_5$

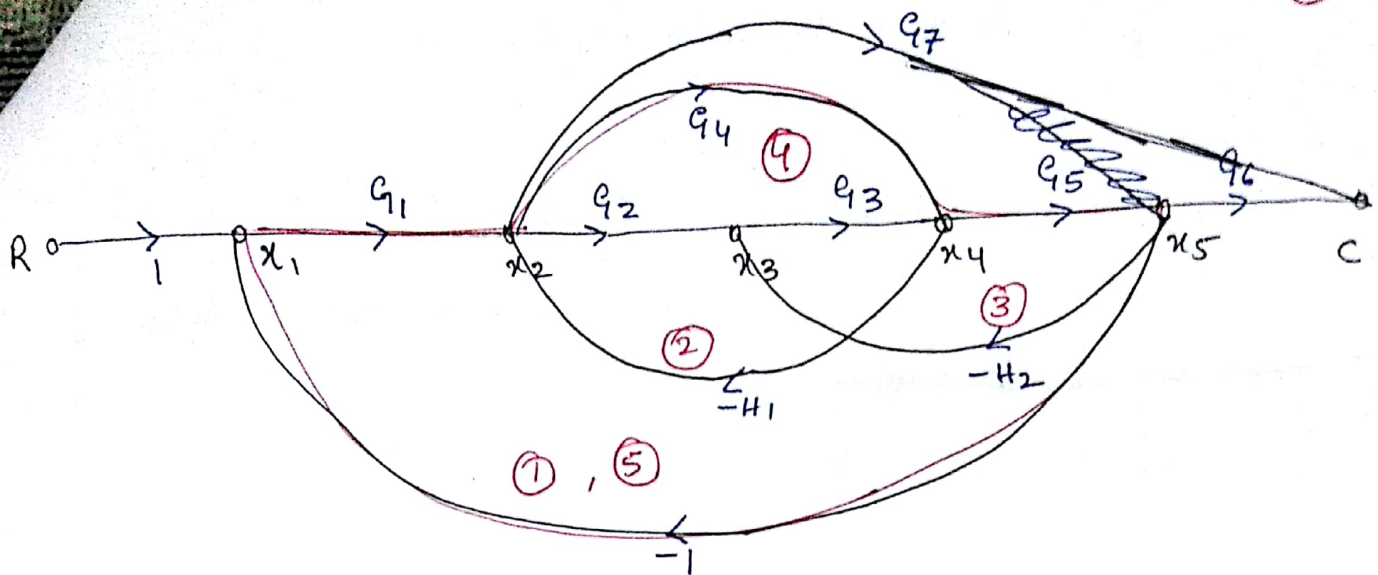
$$x_2 = G_1 x_1 - H_1 x_4$$

$$x_3 = G_2 x_2 - H_2 x_5$$

$$x_4 = G_3 x_3 + G_4 x_2$$

$$x_5 = G_5 x_4$$

$$C = G_6 x_5 + G_7 x_2$$



① There are 3 forward paths with path gains.

$$M_1 = G_1 G_2 G_3 G_5 G_6 \rightarrow R x_1 x_2 x_3 x_4 x_5 C$$

$$M_2 = G_1 G_4 G_5 G_6 \rightarrow R x_1 x_2 x_4 x_5 C$$

$$M_3 = G_1 G_7 \rightarrow R x_1 x_2 C$$

② There are five loops with loop gains.

$$P_{11} = -G_1 G_2 G_3 G_5 \rightarrow x_1 x_2 x_3 x_4 x_5 x_1$$

$$P_{21} = -G_2 G_3 H_1 \rightarrow x_2 x_3 x_4 x_2$$

$$P_{31} = -G_3 G_5 H_2 \rightarrow x_3 x_4 x_5 x_3$$

$$P_{41} = -G_4 H_1 \rightarrow x_2 x_4 x_2$$

$$P_{51} = -G_1 G_4 G_5 \rightarrow x_1 x_2 x_4 x_5 x_1$$

③ There are no two non-touching loops $\therefore P M_2 = 0$

④ F.P. ① is in touch with all loops $\therefore \Delta_1 = 1$

⑤ F.P. ② is in touch with all the loops $\therefore \Delta_2 = 1$

⑥ F.P. ③ is not in touch with $x_3 x_4 x_5 x_3$ loop

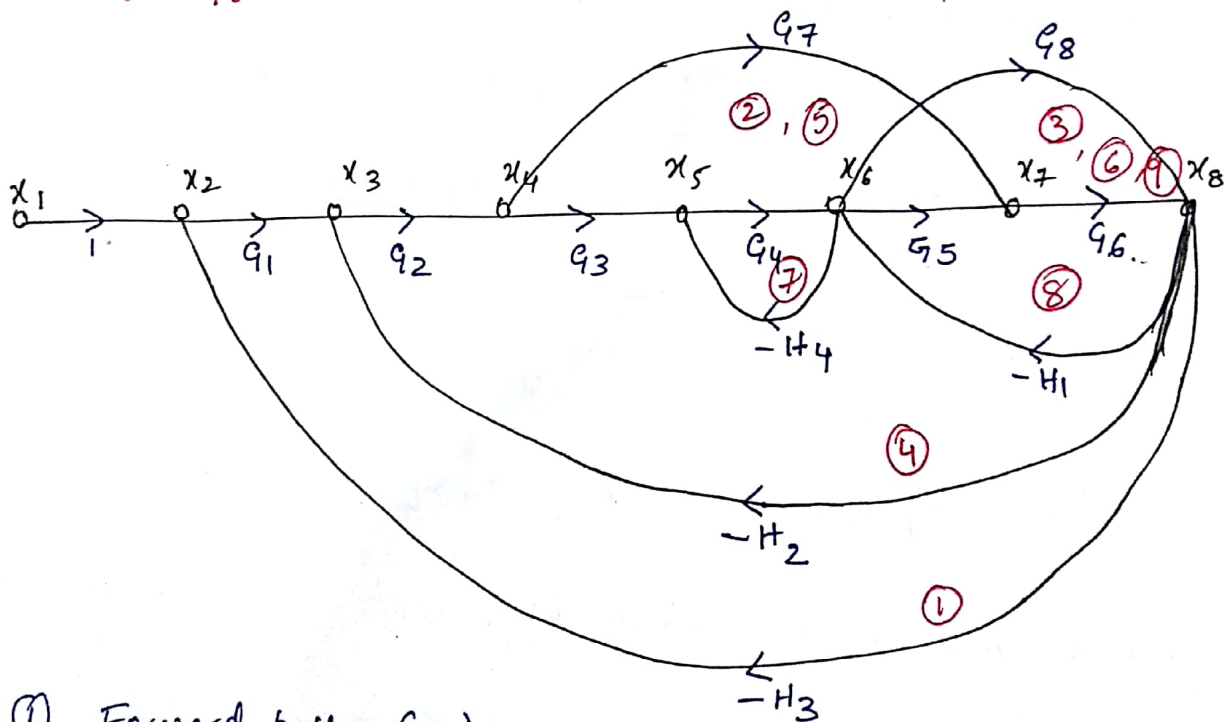
$$\therefore \Delta_3 = 1 - (-G_3 G_5 H_2) = 1 + G_3 G_5 H_2$$

$$\therefore T = \frac{M_1 \Delta_1 + M_2 \Delta_2 + M_3 \Delta_3}{\Delta}$$

$$\therefore T = \frac{G_1 G_2 G_3 G_5 G_6 + G_1 G_4 G_5 G_6 + G_1 G_7 (1 + G_3 G_5 H_2)}{1 + G_1 G_2 G_3 G_5 + G_2 G_3 H_1 + G_3 G_5 H_2 + G_4 H_1 + G_1 G_4 G_5}$$

⑤

$$\begin{aligned} x_2 &= x_1 - H_3 x_8 \\ x_3 &= G_1 x_2 - H_2 x_8 \\ x_4 &= G_2 x_3 \\ x_5 &= G_3 x_4 - H_4 x_6 \\ x_6 &= G_4 x_5 - H_1 x_8 \\ x_7 &= G_7 x_4 + G_5 x_6 \\ x_8 &= G_8 x_6 + G_6 x_7 \end{aligned}$$



① Forward paths (3)

Path	Gain
① $x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8$	$M_1 = G_1 G_2 G_3 G_4 G_5 G_6$
② $x_1 \ x_2 \ x_3 \ x_4 \ x_7 \ x_8$	$M_2 = G_1 G_2 G_7 G_6$
③ $x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_8$	$M_3 = G_1 G_2 G_3 G_4 G_8$

closed loops

(7)

closed loop	loop gain	
① $x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_2$	$-G_1 G_2 G_3 G_4 G_5 G_6 H_3$	P_{11}
② $x_2 x_3 x_4 x_7 x_8 x_2$	$-G_1 G_2 G_7 G_6 H_3$	P_{21}
③ $x_2 x_3 x_4 x_5 x_6 x_8 x_2$	$-G_1 G_2 G_3 G_4 G_8 H_3$	P_{31}
④ $x_3 x_4 x_5 x_6 x_7 x_8 x_3$	$-G_2 G_3 G_4 G_5 G_6 H_2$	P_{41}
⑤ $x_3 x_4 x_7 x_8 x_3$	$-G_2 G_7 G_6 H_2$	P_{51}
⑥ $x_3 x_4 x_5 x_6 x_8 x_3$	$-G_2 G_3 G_4 G_8 H_2$	P_{61}
⑦ $x_5 x_6 x_5$	$-G_4 H_4$	P_{71}
⑧ $x_6 x_7 x_8 x_6$	$-G_5 G_6 H_1$	P_{81}
⑨ $x_6 x_8 x_6$	$-G_8 H_1$	G_{91}

P_{M1}

Two non-touching loops	product gain
$x_2 x_3 x_4 x_7 x_8 x_2$ & $x_5 x_6 x_5$	$(G_1 G_2 G_7 G_6 (-H_3)) (-G_4 H_4)$
	$= G_1 G_2 G_4 G_6 G_7 H_3 H_4 \rightarrow P_{21}$
$x_3 x_4 x_7 x_8 x_3$ & $x_5 x_6 x_5$	$(G_2 G_7 G_6 (-H_2)) (-G_4 H_4)$
	$= G_2 G_6 G_7 G_4 H_2 H_4 \rightarrow P_{22}$

P_{M2}

④ FP-① is in touch with all the loops $\therefore \Delta_1 = 1$

FP-② is not in touch with $x_5 x_6 x_5$ loop $\therefore \Delta_2 = 1 - (-G_4 H_4)$
 $= \Delta_2 = 1 + G_4 H_4$

FP-③ is not in touch with all the loops.

$\therefore \Delta_3 = 1$

$$T = \frac{M_1 \Delta_1 + M_2 \Delta_2 + M_3 \Delta_3}{\Delta}$$

where $\Delta = 1 - P_{M1} + P_{M2}$

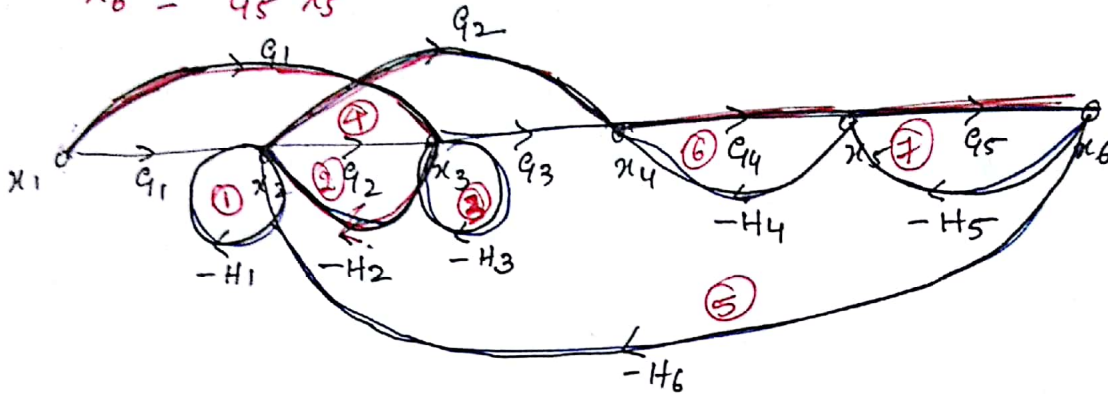
$$⑥ \quad x_2 = G_1 x_1 - H_1 x_2 - H_2 x_3 - H_6 x_6$$

$$x_3 = G_1 x_1 + G_2 x_2 - H_3 x_3$$

$$x_4 = G_2 x_2 + G_3 x_3 - H_4 x_5$$

$$x_5 = G_4 x_4 - H_5 x_6$$

$$x_6 = G_5 x_5$$



①

Forward path.

Forward gain.

FP① $\Rightarrow x_1 x_2 x_3 x_4 x_5 x_6$

$G_1 G_2 G_3 G_4 G_5 = M_1$

FP② $\Rightarrow x_1 x_3 x_4 x_5 x_6$

$G_1 G_3 G_4 G_5 = M_2$

FP③ $\Rightarrow x_1 x_3 x_2 x_4 x_5 x_6$

$-G_1 H_2 G_2 G_4 G_5 = M_3$

FP④ $\Rightarrow x_1 x_2 x_4 x_5 x_6$

$G_1 G_2 G_4 G_5$

②

closed loop

loop gain.

① $x_2 x_2$

$-H_1 \rightarrow P_{11}$

② $x_2 x_3 x_2$

$-G_2 H_2 \rightarrow P_{21}$

③ $x_3 x_3$

$-H_3 \rightarrow P_{31}$

④ $x_2 x_3 x_4 x_5 x_6 x_2$

$-G_2 G_3 G_4 G_5 H_6 \rightarrow P_{41}$

⑤ $x_2 x_4 x_5 x_6 x_2$

$-G_2 G_4 G_5 H_6 \rightarrow P_{51}$

⑥ $x_4 x_5 x_4$

$-G_4 H_4 \rightarrow P_{61}$

⑦ $x_5 x_6 x_5$

$-G_5 H_5 \rightarrow P_{71}$

ΣP_{M1}

(8)

Two non touching loops	Product Gain.
① 1 & 3.	$H_1 H_3 \rightarrow P_{12}$
② 1 & 6	$G_5 H_4 H_1 \rightarrow P_{22}$
③ 1 & 7	$G_5 H_5 H_1 \rightarrow P_{32}$
④ 2 & 6	$G_2 G_4 H_2 H_4 \rightarrow P_{42}$
⑤ 2 & 7	$G_2 G_5 H_2 H_5 \rightarrow P_{52}$
⑥ 3 & 5.	$G_2 G_4 G_5 H_3 H_6 \rightarrow P_{62}$
⑦ 3 & 6	$G_4 H_4 H_3 \rightarrow P_{72}$
⑧ 3 & 7	$G_5 H_3 H_5 \rightarrow P_{82}$

} $\sum P_{M2}$

④

Three non touching loops	Product Gain.
① 1, 3 & 6	$-H_1 H_3 G_4 H_4 \rightarrow P_{13}$
② 1, 3 & 7	$-H_1 H_3 G_5 H_5 \rightarrow P_{23}$

} $\sum P_{M3}$

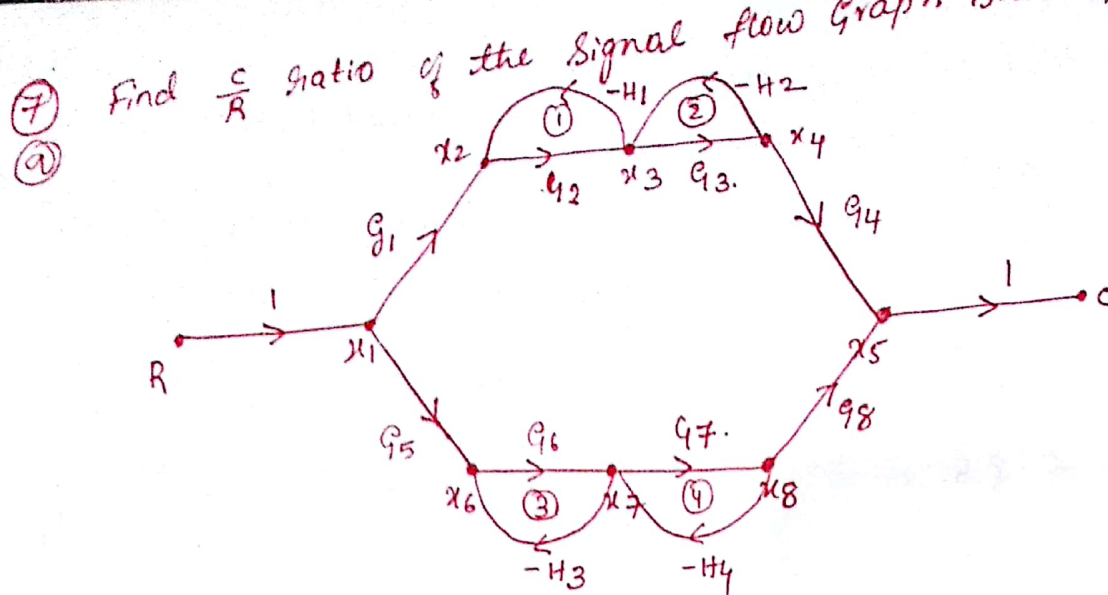
⑤ $FP_{①}$ is in touch with all the loops $\therefore \Delta_1 = 1$

⑥ $FP_{②}$ is not in touch with H_1 loop $\therefore \Delta_2 = 1 - (-H_1)$
 $= 1 + H_1$

⑦ $FP_{③}$ is in touch with all the loops $\therefore \Delta_3 = 1$

⑧ $FP_{④}$ is not in touch with H_3 loop $\therefore \Delta_4 = 1 - (-H_3)$
 $= 1 + H_3$

$$T = \frac{M_1 \Delta_1 + M_2 \Delta_2 + M_3 \Delta_3 + M_4 \Delta_4}{1 - \sum P_{M1} + \sum P_{M2} - \sum P_{M3}}$$



① Forward path | forward gain.

FP ① $R \rightarrow x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4 \rightarrow x_5 \rightarrow C$ $M_1 = G_1 G_2 G_3 G_4$

FP ② $R \rightarrow x_1 \rightarrow x_6 \rightarrow x_7 \rightarrow x_8 \rightarrow x_5 \rightarrow C$ $M_2 = G_5 G_6 G_7 G_8$

② There are 4 closed loops.

$P_{11} = -G_2 H_1$, $P_{21} = -G_3 H_2$, $P_{31} = -G_6 H_3$ & $P_{41} = -G_7 H_4$.

③ There are 4, two non touching loops with product gain.

$P_{12} = G_2 G_6 H_1 H_3$; $P_{22} = G_2 G_7 H_1 H_4$.

$P_{32} = G_3 G_6 H_2 H_3$; $P_{42} = G_3 G_7 H_2 H_4$

④ There are no 3 non touching loops. $\therefore P_{M3} = 0$

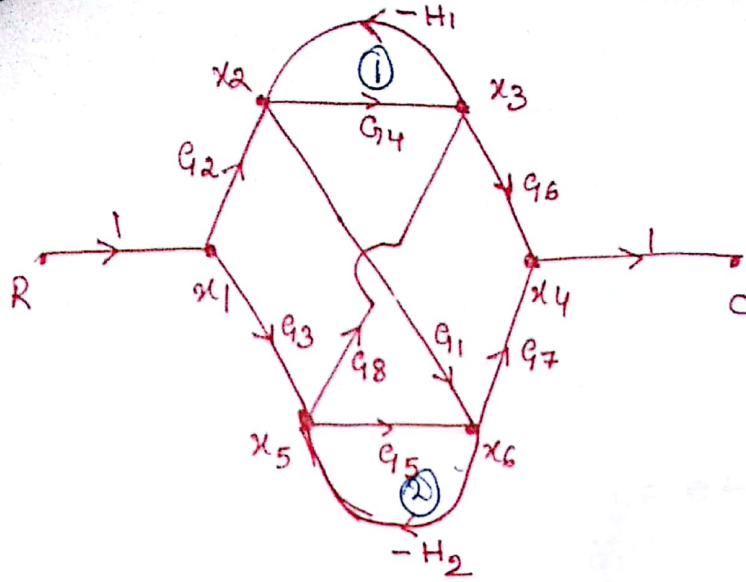
⑤ FP ① is not in touch with ③ & ④.

$\therefore \Delta_1 = 1 - [(-G_6 H_3) + (-G_7 H_4)] = 1 + G_6 H_3 + G_7 H_4$

FP ② is not in touch with ① & ②

$\therefore \Delta_2 = 1 - [(-G_2 H_1) + (-G_3 H_2)] = 1 + G_2 H_1 + G_3 H_2$

$\therefore T = \frac{M_1 \Delta_1 + M_2 \Delta_2}{1 - P_{M1} + P_{M2}}$



① Forward paths.	Forward gain
FP① $R \rightarrow x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4 \rightarrow C$	$M_1 = G_2 G_4 G_6$
FP② $R \rightarrow x_1 \rightarrow x_5 \rightarrow x_6 \rightarrow x_4 \rightarrow C$	$M_2 = G_3 G_5 G_7$
FP③ $R \rightarrow x_1 \rightarrow x_5 \rightarrow x_3 \rightarrow x_4 \rightarrow C$	$M_3 = G_3 G_8 G_6$
FP④ $R \rightarrow x_1 \rightarrow x_2 \rightarrow x_6 \rightarrow x_4 \rightarrow C$	$M_4 = G_2 G_1 G_7$
FP⑤ $R \rightarrow x_1 \rightarrow x_5 \rightarrow x_3 \rightarrow x_2 \rightarrow x_6 \rightarrow x_4 \rightarrow C$	$M_5 = -G_3 G_8 H_1 G_1 G_7$
FP⑥ $R \rightarrow x_1 \rightarrow x_2 \rightarrow x_6 \rightarrow x_5 \rightarrow x_3 \rightarrow x_4 \rightarrow C$	$M_6 = -G_2 G_1 H_2 G_8 G_6$
② closed loop	loop gain.
① $x_2 \rightarrow x_3 \rightarrow x_2$	$-G_4 H_1$
② $x_5 \rightarrow x_6 \rightarrow x_5$	$-G_5 H_2$
③ $x_2 \rightarrow x_6 \rightarrow x_5 \rightarrow x_3 \rightarrow x_2$	$G_1 G_8 H_1 H_2$
$\left. \begin{matrix} -G_4 H_1 \\ -G_5 H_2 \\ G_1 G_8 H_1 H_2 \end{matrix} \right\} P_{M1}$	
③ Two non touching loops	Product gain
① ④ ②	$G_4 G_5 H_1 H_2$

④ FP-① is not in touch with loop ②
 $\therefore \Delta_1 = 1 - (-G_5 H_2) = 1 + G_5 H_2$.

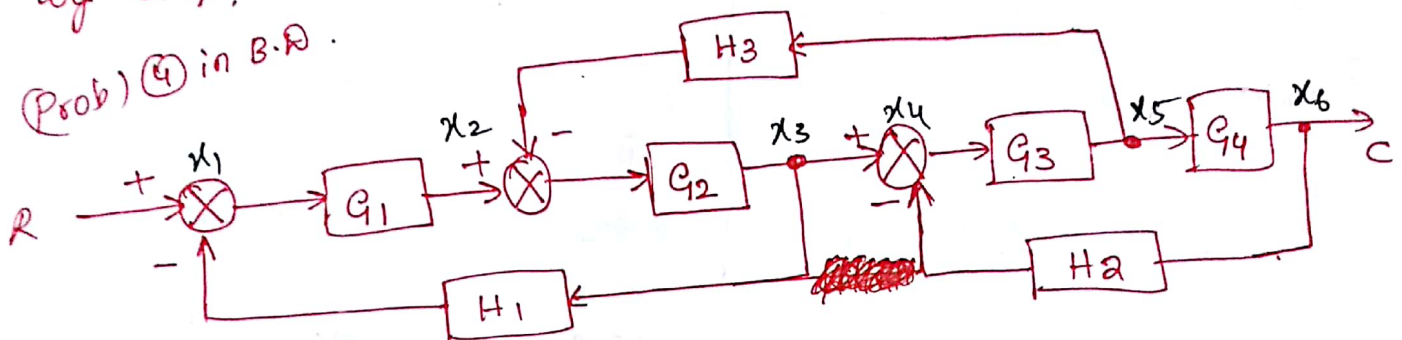
FP-② is not in touch with loop ①
 $\therefore \Delta_2 = 1 - (-G_4 H_1) = 1 + G_4 H_1$

$\Delta_3 = \Delta_4 = \Delta_5 = \Delta_6 = 1$ $\therefore$ all these FP's are in touch with all loops.

$$\therefore T = \frac{M_1 \Delta_1 + M_2 \Delta_2 + M_3 \Delta_3 + M_4 \Delta_4 + M_5 \Delta_5 + M_6 \Delta_6}{1 - \sum PM_1 + \sum PM_2}$$

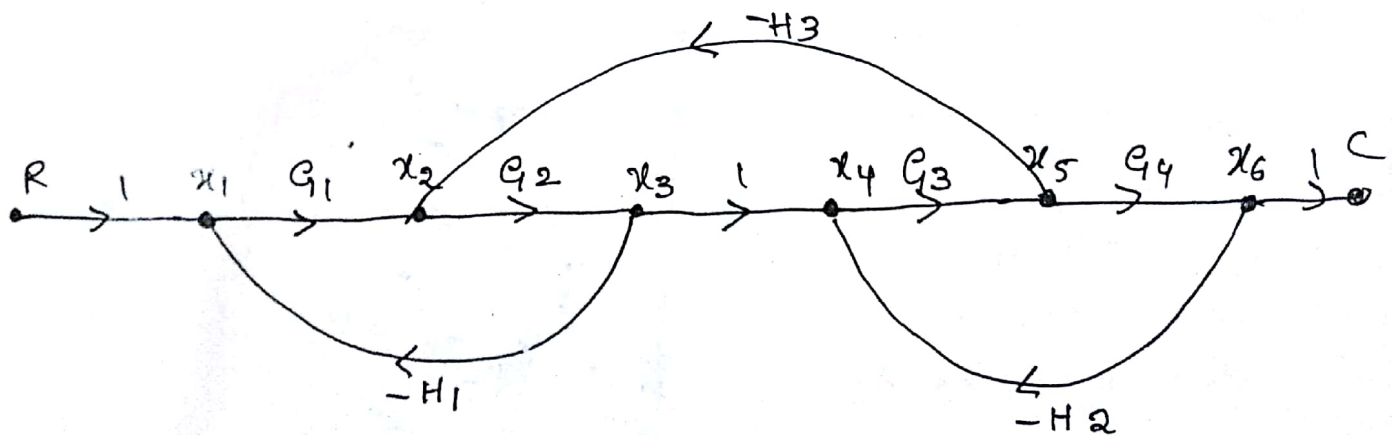
⑧ Determine the TF of the B.D shown and verify the same by SFG.

(Prob) ④ in B.D.



$$\frac{C}{R} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 + G_2 G_3 H_3 + G_1 G_2 H_1 + G_1 G_2 H_1 H_2 G_3 G_4}$$

SFG



$$M_1 = G_1 G_2 G_3 G_4$$

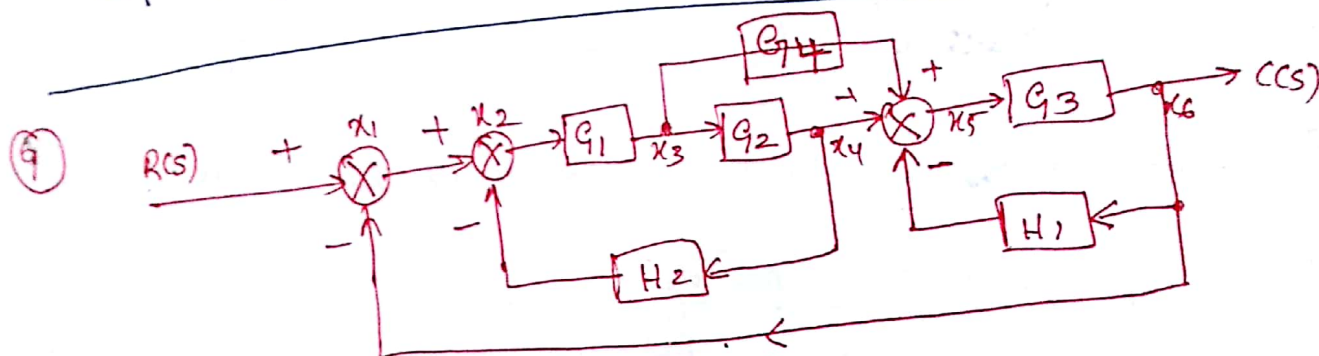
closed loop	loop gain
$x_1 x_2 x_3 x_1$	$-G_1 G_2 H_1$
$x_4 x_5 x_6 x_4$	$-G_3 G_4 H_2$
$x_2 x_3 x_4 x_5 x_2$	$-G_2 G_3 H_3$

Two non touching loops	product gain
1 & 2	$G_1 G_2 G_3 G_4 H_1 H_2$

$\Delta_1 = 1$ (Because all loop touch FP ①)

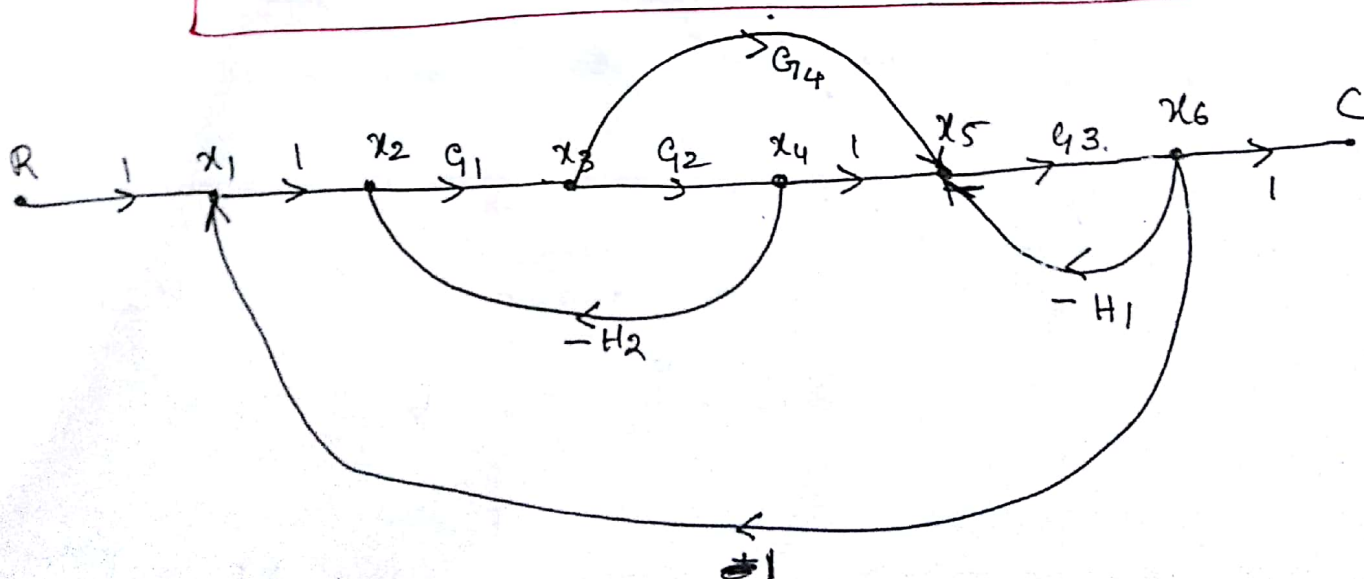
$$T = \frac{M_1 \Delta_1}{\Delta}$$

$$T = \frac{G_1 G_2 G_3 G_4 \cdot 1}{1 + G_1 G_2 H_1 + G_3 G_4 H_2 + G_2 G_3 H_3 + G_1 G_2 G_3 G_4 H_1 H_2}$$

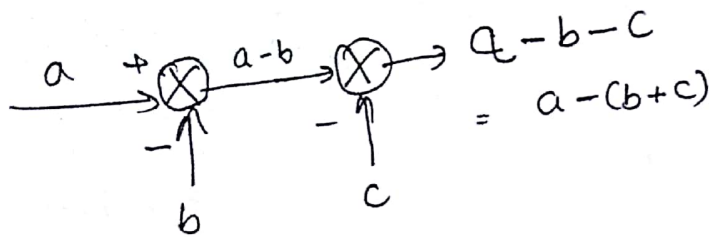


(Refer Prob ⑤ under BD Redn. whose ans is

$$TF = \frac{G_1 G_2 G_3 + G_1 G_3 G_4}{1 + H_1 G_3 + G_1 G_2 H_2 + G_1 G_2 H_1 H_2 G_3 + G_1 G_2 G_3 + G_1 G_3 G_4}$$



(Associative law)

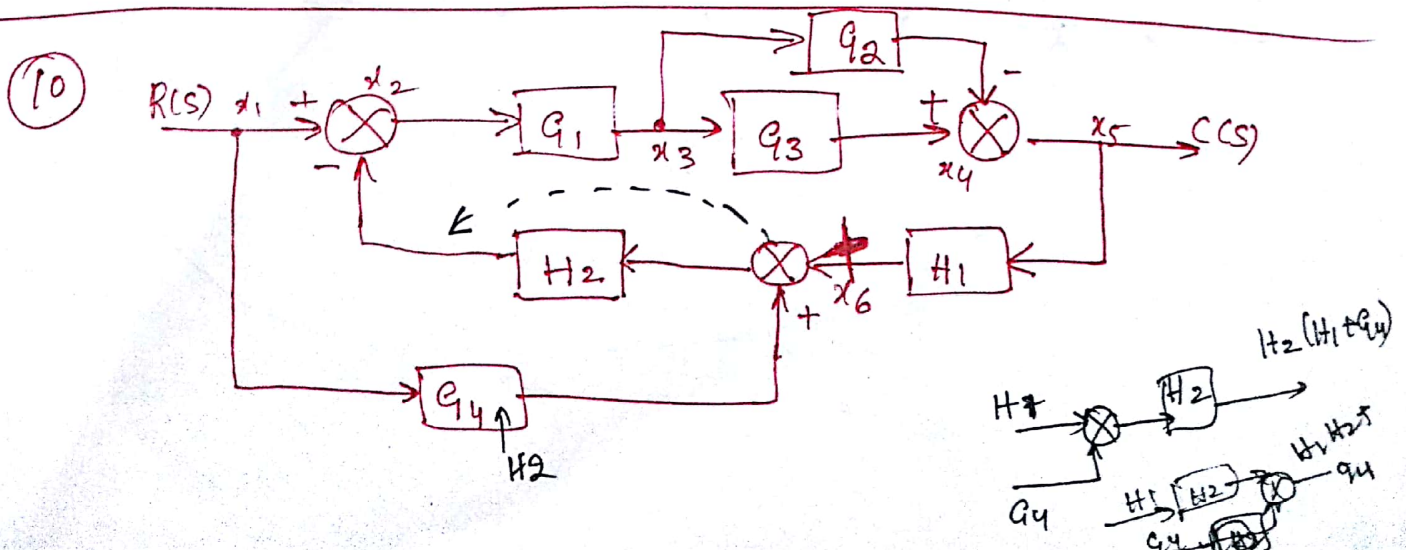


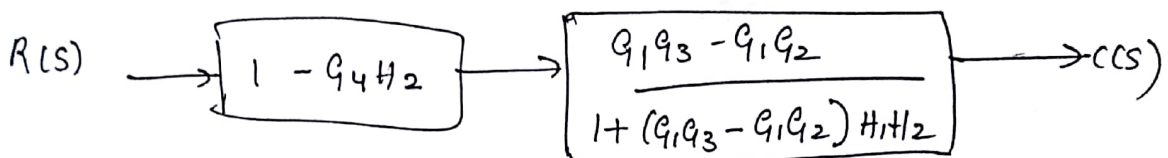
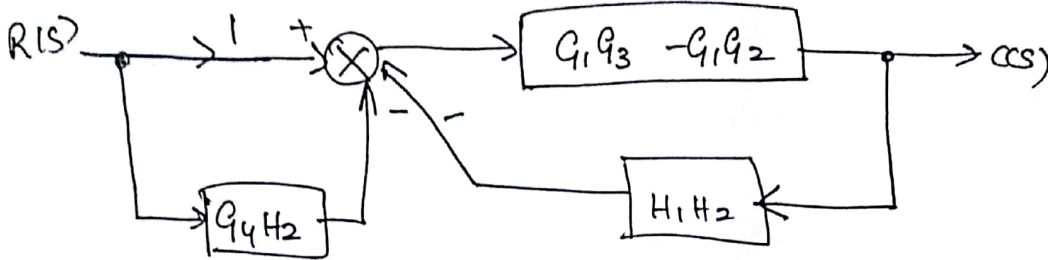
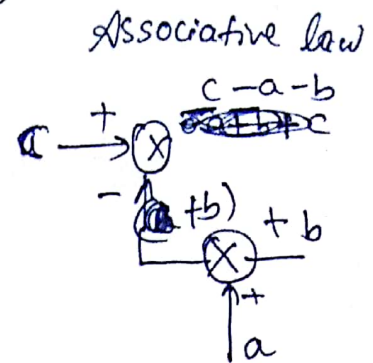
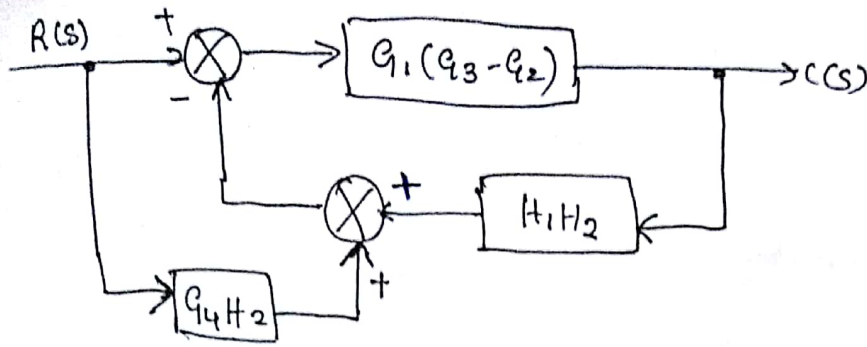
Note: The nodes x_1 & x_2 can be combined as one single node ($\because$ associative). where as x_4 & x_5 can't be combined.

- ① $M_1 = G_1 G_2 G_3$ & $M_2 = G_1 G_4 G_3$
- ② $P_{11} = -G_1 G_2 H_2$ $P_{21} = -G_1 G_4 G_3$ $P_{31} = -G_1 G_2 G_3$ $P_{41} = -G_3 H_1$
- ③ 2 non-touching loops. $P_{12} = G_1 G_2 H_2 G_3 H_1$ (P_{11} & P_{41})
- ④ $\Delta_1 = 1$ & $\Delta_2 = 1$ Both FP touch all the loops.
- ⑤
$$T = \frac{M_1 \Delta_1 + M_2 \Delta_2}{1 - \sum P_{M1} + \sum P_{M2}}$$

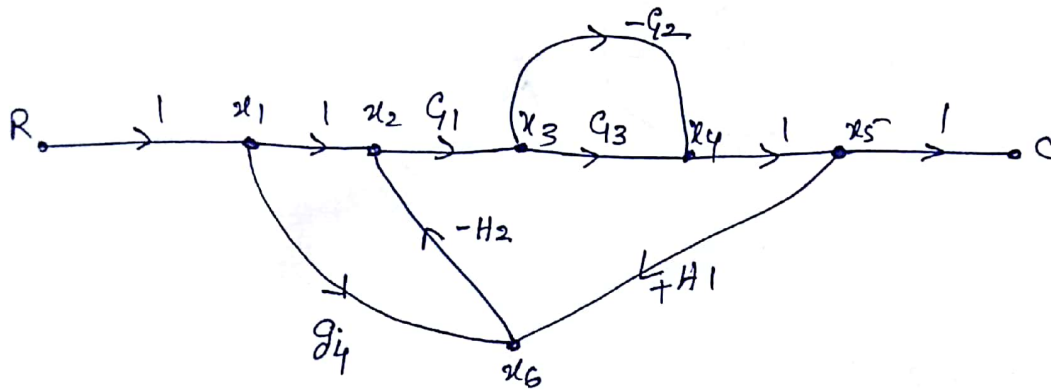
$$T = \frac{G_1 G_2 G_3 + G_1 G_4 G_3}{1 + G_1 G_2 H_2 + G_1 G_4 G_3 + G_1 G_2 G_3 + G_3 H_1 + G_1 G_2 G_3 H_1 H_2}$$

Thus both are proved to be similar.





$$\frac{C(s)}{R(s)} = \frac{G_1Q_3 - G_1Q_2 - G_1Q_3G_4H_2 + G_1Q_2G_4H_2}{1 + G_1Q_3H_1H_2 - G_1Q_2H_1H_2}$$



① $M_1 = G_1Q_3$ $M_2 = -G_1Q_2$ $M_3 = -G_4G_1Q_3H_2$ $M_4 = G_4G_1Q_2H_2$

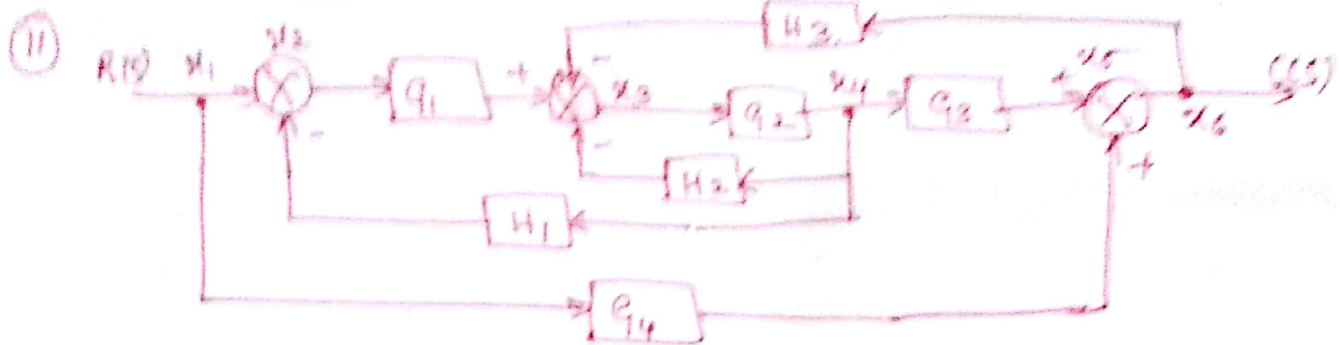
② Loops

$P_{11} = -G_1Q_3H_1H_2$ $P_{21} = G_1Q_2H_1H_2$

③ There are no two non touching loops i.e. $P_{m2} = 0$

$\Delta_1 = 1$ $\Delta_2 = 1$ $\Delta_3 = 1$ & $\Delta_4 = 1$

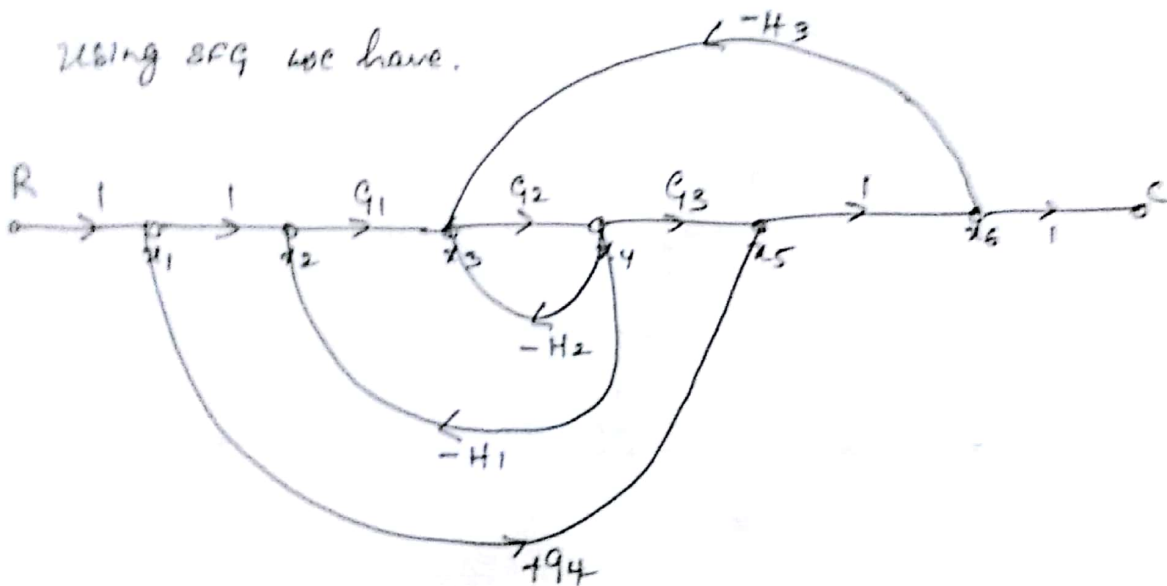
$$T = \frac{G_1 G_3 - G_1 G_2 - G_1 G_3 G_4 H_2 + G_1 G_2 G_4 H_2}{1 + G_1 G_3 H_1 H_2 - G_1 G_2 H_1 H_2}$$



(Refer Prob No (8) under BD)

$$TF = \frac{G_1 G_2 G_3 + G_2 G_4 H_2 + G_1 G_2 G_4 H_1 + G_4}{1 + G_2 H_2 + G_1 G_2 H_1 + G_2 G_3 H_3}$$

using SFG we have.



① $M_1 = G_1 G_2 G_3$ & $M_2 = G_4$

② $P_{11} = -G_1 G_2 H_1$; $P_{21} = -G_2 H_2$; $P_{31} = -G_2 G_3 H_3$

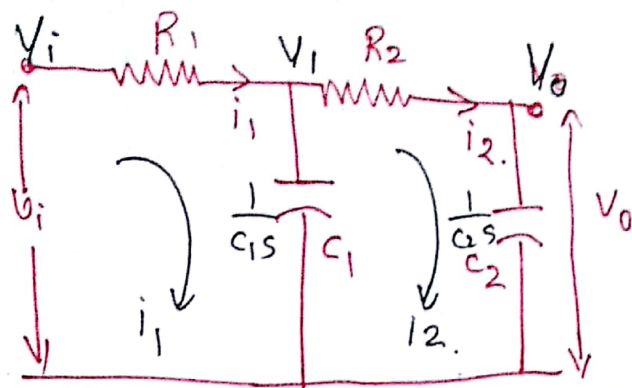
③ $\Delta_1 = 1$; $\Delta_2 = 1 + (-G_2 H_2 - G_1 G_2 H_1)$

$$\therefore TF = \frac{G_1 G_2 G_3 \cdot 1 + G_4 (1 + G_2 H_2 + G_1 G_2 H_1)}{1 + G_2 H_2 + G_1 G_2 H_1 + G_2 G_3 H_3}$$

$$TF = \frac{G_1 G_2 G_3 + G_4 + G_4 G_2 H_2 + G_4 G_1 G_2 H_1}{1 + G_2 H_2 + G_1 G_2 H_1 + G_2 G_3 H_3}$$

construct B.D and also draw SFQ to determine their TF.

(12)

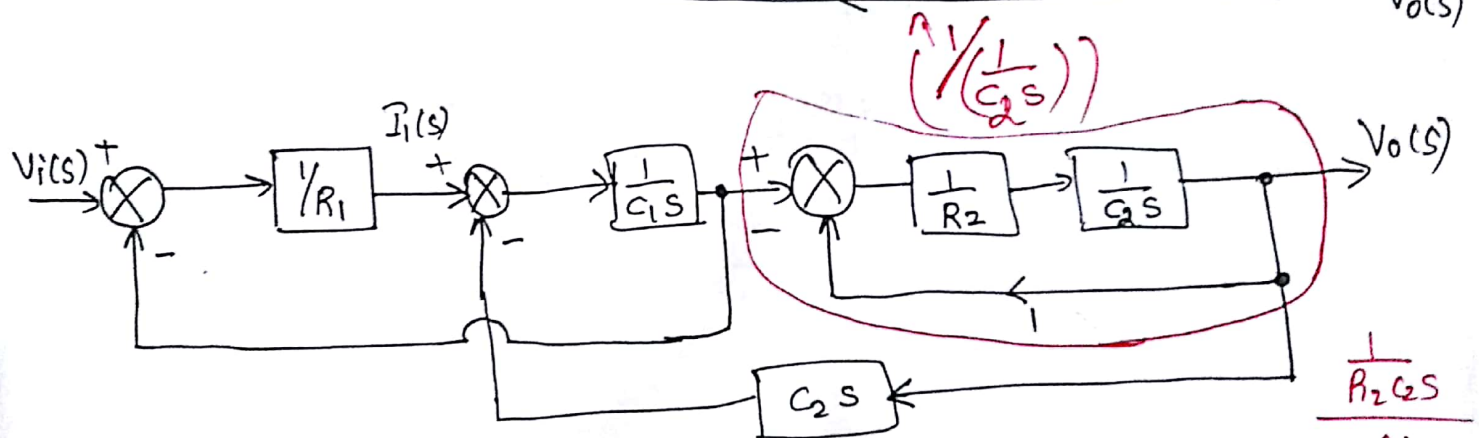
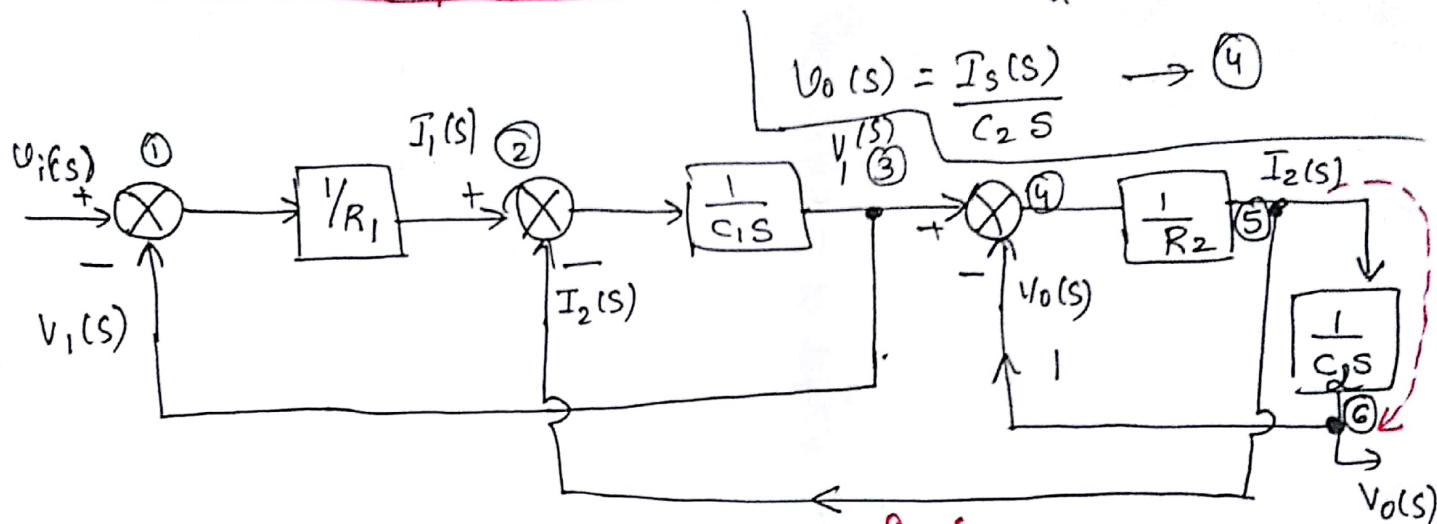


$$I_1(s) = \frac{V_i(s) - V_1(s)}{R_1} \rightarrow (1)$$

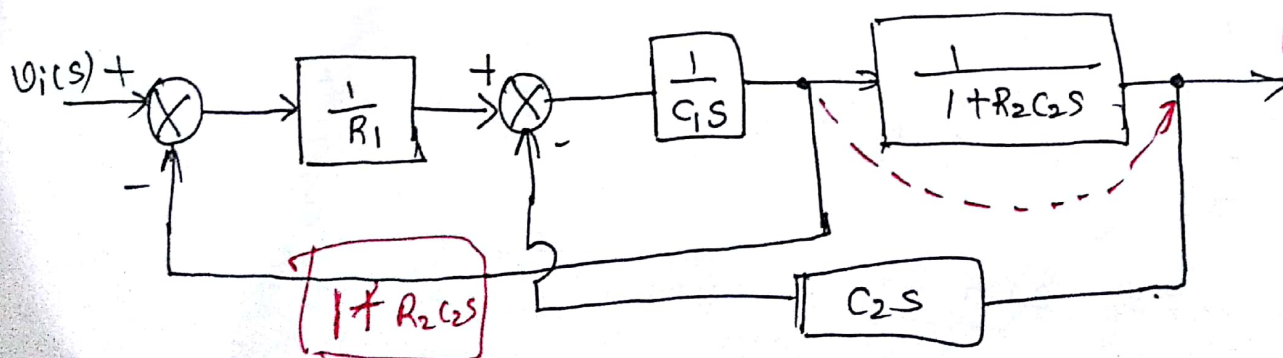
$$V_1(s) = [I_1(s) - I_2(s)] \frac{1}{C_1 s} \rightarrow (2)$$

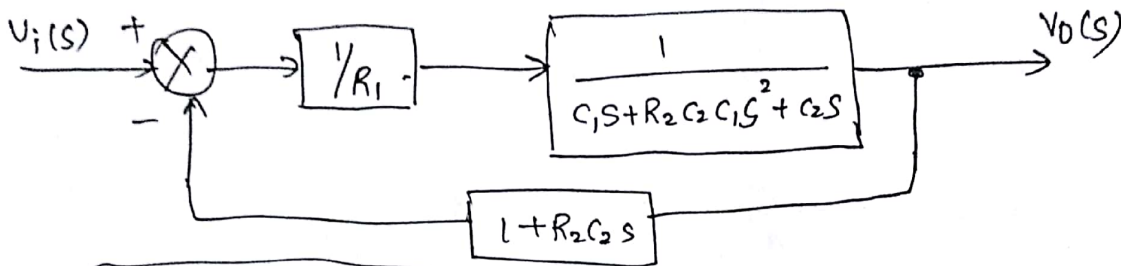
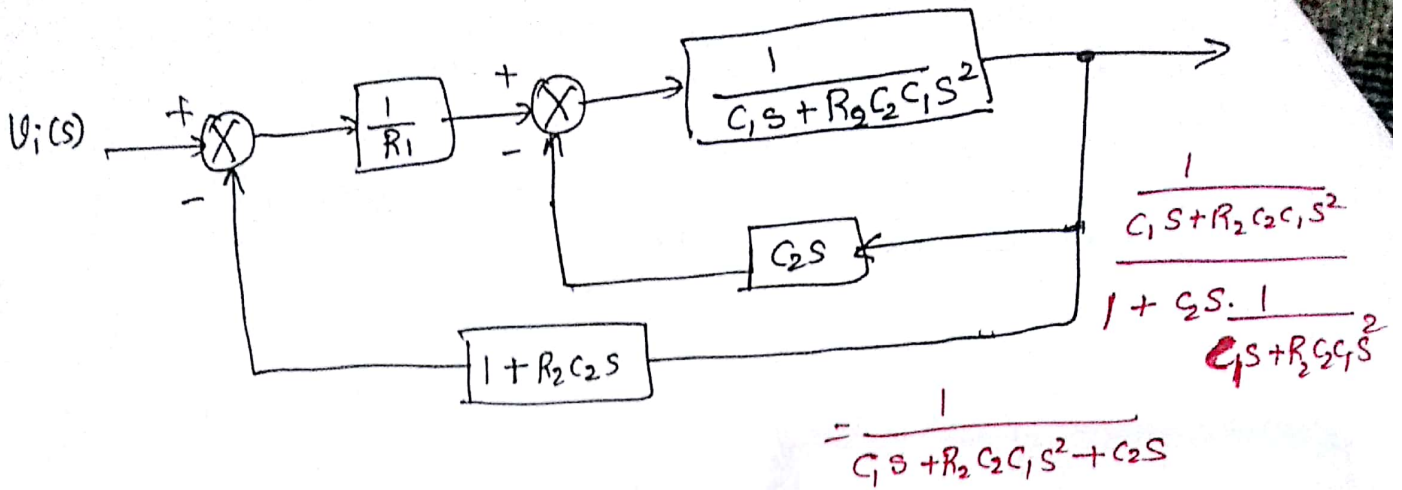
$$I_2(s) = \frac{V_1(s) - V_o(s)}{R_2} \rightarrow (3)$$

$$V_o(s) = \frac{I_2(s)}{C_2 s} \rightarrow (4)$$



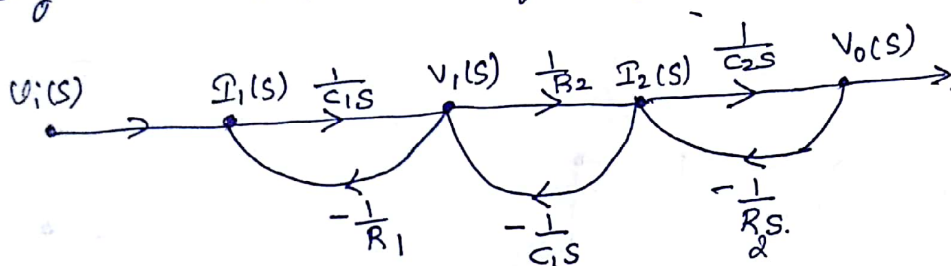
$$\frac{1}{R_2 C_2 s} \cdot \frac{1}{1 + 1 \cdot \left(\frac{1}{R_2 C_2 s} \right)} = \frac{1}{1 + R_2 C_2 s}$$





$$\frac{V_o(s)}{U_i(s)} = \frac{1}{R_1 C_1 s + R_2 R_1 C_1 C_2 s^2 + R_1 C_2 s + 1 + R_2 C_2 s}$$

Signal flow Graph from Equations



$$M_1 = \frac{1}{R_1 R_2 C_1 C_2 s^2}$$

$$P_{11} = -\frac{1}{C_1 R_1 s} \quad P_{21} = -\frac{1}{C_1 R_2 s} \quad P_{31} = -\frac{1}{C_2 R_2 s}$$

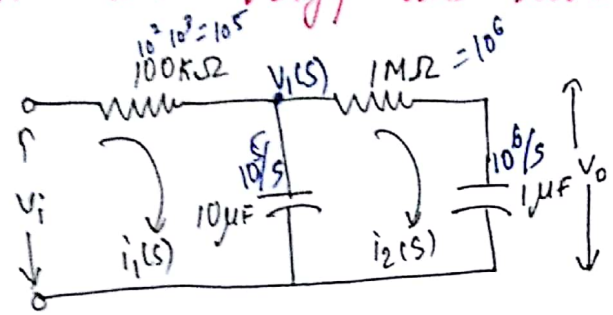
$$P_{12} = (P_{11} \text{ \& } P_{31}) = \frac{1}{R_1 C_1 s^2 R_2 C_2} \Rightarrow \Delta_1 = 1$$

$$\therefore TF = \frac{\sum M_k \Delta_k}{\Delta} = \frac{1}{R_1 R_2 C_1 C_2 s^2} \cdot \frac{1}{1 + \frac{1}{C_1 R_1 s} + \frac{1}{C_1 R_2 s} + \frac{1}{C_2 R_2 s} + \frac{1}{R_1 C_1 s^2 R_2 C_2}}$$

$$TF = \frac{1}{R_1 R_2 C_1 C_2 s^2} \cdot \frac{1}{(R_1 C_1 s^2 R_2 C_2 + R_2 C_2 s + R_1 C_2 s + C_1 R_1 s + 1) / R_1 C_1 s^2 R_2 C_2}$$

$$R_1 R_2 C_2 C_1 S^2 + R_2 C_2 S + R_1 C_2 S + R_1 C_1 S + 1$$

(13) For the N/w shown, draw the SFG and determine the TF and verify the result from block diagram

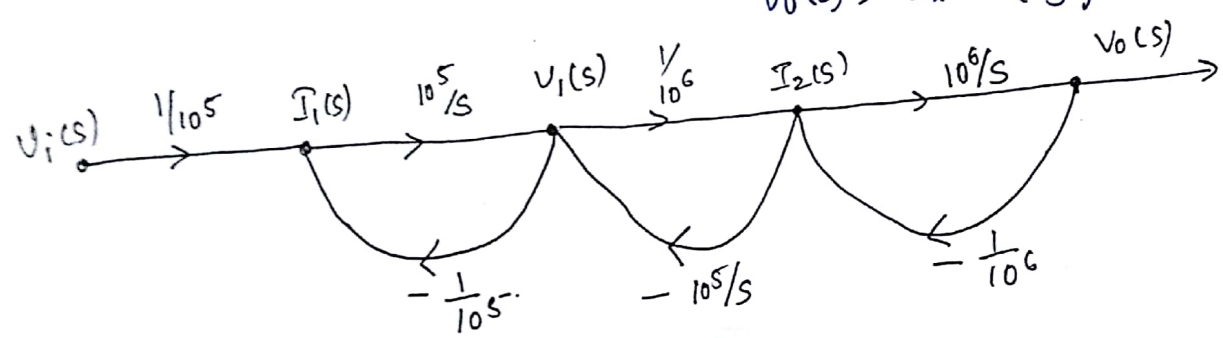


$$I_1(s) = \frac{V_i(s) - V_1(s)}{10^5} \rightarrow (1)$$

$$V_1(s) = (I_1(s) - I_2(s)) \frac{10^5}{s} \rightarrow (2)$$

$$I_2(s) = \frac{V_1(s) - V_o(s)}{10^6} \rightarrow (3)$$

$$V_o(s) = I_2(s) \cdot \left(\frac{10^6}{s}\right) \rightarrow (4)$$



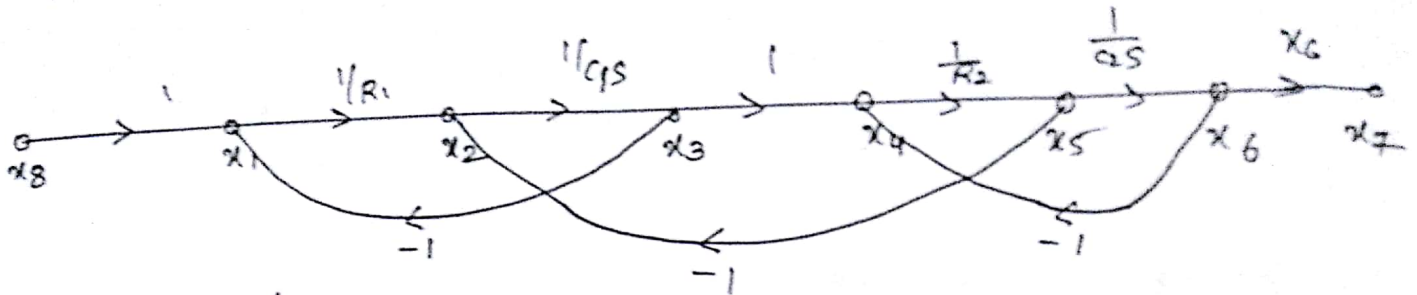
$$M_1 = \frac{1}{10^5} \times \frac{10^5}{s} \times \frac{1}{10^6} \times \frac{10^6}{s} = \frac{1}{s^2}$$

$$P_{11} = -\frac{1}{s} \quad P_{12} = -\frac{1}{10s} \quad P_{13} = -\frac{1}{s}$$

$$P_{21} = P_{11} \text{ \& } P_{13} = \frac{1}{s^2}, \quad \Delta_1 = 1$$

$$TF = \frac{\frac{1}{s^2}}{1 + \frac{1}{s} + \frac{1}{10s} + \frac{1}{s^2} + \frac{1}{s}} = \frac{\frac{1/s^2}{10s^2}}{\frac{10s^2 + 10s + s + 10s + 10}{10s^2}} = \frac{10}{10s^2 + 21s + 10}$$

OR from B.D.



$$P_1 = \frac{1}{R_1 R_2 C_1 C_2 S^2} \Rightarrow \text{FP - ①}$$

$$P_{11} = -\frac{1}{R_1 C_1 S} \quad P_{21} = -\frac{1}{C_1 R_2 S} \quad P_{31} = -\frac{1}{R_2 C_2 S}$$

$$P_{12} = P_{11} \text{ \& } P_{31} = \frac{1}{R_1 C_1 S} \cdot \frac{1}{R_2 C_2 S} \quad \Delta_1 = 1$$

This is ill^{er} to drawing from eqn's